Yevhen Perehuda

PhD student in Cryptography

44799, Bochum, Germany
+49 157 53303413
yevhen.perehuda@rub.de

Profile Summary

PhD researcher at Ruhr University Bochum, working on the mathematical foundations of security arguments and cryptanalysis. Published at **CRYPTO 2025** and **EUROCRYPT 2026**; develops practical analysis tooling in CUDA, C, and Python. Expected PhD defense: **March 2027** (could be adjusted by a few months if necessary).

Personal Details

Nationality	Ukraine

Research & Professional Experience

Apr. 2026 - May. 2026	**Visiting Scholar** at the KU Leuven, Belgium
Oct. 2023 - Present	**PhD student** in the Symmetric Cryptography Group, Ruhr University Bochum, Germany
Research group	Prof. Gregor Leander
Research project	Research in the mathematical foundations of security arguments and cryptanalysis for symmetric cryptographic primitives. Implementing experiments and automatic tools in CUDA, C++ and Python, including SAT/MILP-based analysis.
Teaching Assistant Experience	○ Boolean Functions (Master) ○ Symmetric Cryptography (Master) ○ Math 1 (Bachelor)
May. 2021 - Aug. 2023	**Student Research Assistant** at the University of Luxembourg, Luxembourg
Project	○ Developing and benchmarking new evolutionary-algorithm techniques in Python; automated test-suite analysis. ○ Support for university research facilitators of grant proposals.
Oct. 2018 - May. 2019	**Python developer** at Momentum Bots, Kyiv, Ukraine
Project	Developing Telegram bots in Python.
Dec. 2017 - Jan. 2021	**Math tutor** for university students (linear algebra, probability)

Education

Jan. 2020 - Sep. 2022	**M.Sc.** *in Information and Computer Science* Computer Science Department, University of Luxembourg, Luxembourg. Supervised by Dr. Andy Rupp. Master thesis on *Integral Attack Resistance in Block Ciphers: New Bounds and Generalized Framework.*
Sep. 2022 - Sep. 2023	**M.Sc.** *in Quantitative Economics and Finance* Department of Finance, University of Luxembourg, Luxembourg. Supervised by Prof. Benjamin Holcblat. Master thesis on *Residual-based CUSUM test in X-ARCH models*
Sep. 2015 - Sep. 2019	**B.Sc.** *in Information and Computer Science* National Technical University of Ukraine, Kyiv. Supervised by Prof. Victor Pasko. Bachelor thesis on *Control system for a positional-type CNC machine tool.*

Publications

- Christof Beierle, Phil Hebborn, Gregor Leander, and Yevhen Perehuda, *Integral Resistance of Block Ciphers with Key Whitening by Modular Addition*, in *CRYPTO 2025*. Available at https://eprint.iacr.org/2025/1050.
- Christof Beierle, Gregor Leander, and Yevhen Perehuda, *When the Wrong Key Lives On: The Key-Recovery Procedure in Integral Attacks*. in *EUROCRYPT 2026*. Available at https://eprint.iacr.org/2025/2011.
- Jens Alich, Thomas Eisenbarth, Hossein Hadipour, Gregor Leander, Felix Mächtle, Yevhen Perehuda, Shahram Rasoolzadeh, Jonas Sander, and Cihangir Tezcan, *Attacks and Remedies for Randomness in AI: Cryptanalysis of PHILOX and THREEFRY* – analysis of counter-based PRNGs used in parallel systems. Manuscript under submission (2025). Available at https://eprint.iacr.org/2025/2161. *Preprint*.

(Sub-)Reviewer

Conferences	*CRYPTO (2024, 2025, 2026), EUROCRYPT(2026)*
Journals	Cryptography and Communications

Talks, Presentation and Organization

Organization
- Workshop on GPU-Tailored Modern Symmetric Cryptography (GEMS 2026), EUROCRYPT 2026 Affiliated event, Rome, Italy, 10 May 2026

Presentations
- "When the Wrong Key Lives On: The Key-Recovery Procedure in Integral Attacks", GelreCrypt workshop, Nijmegen, Netherlands, Nov. 2025
- "Attacks and Remedies for Randomness in AI: Cryptanalysis of PHILOX and THREEFRY", Radboud Digital Security group Lunch Talk, Nijmegen, Netherlands, 4 March 2026
- "Attacks and Remedies for Randomness in AI: Cryptanalysis of PHILOX and THREEFRY", Cryptography seminar, Rennes, France, 13 March 2026

Referees

Prof. Dr. Gregor Leander	Chair of Symmetric Cryptography, Faculty of Computer Science, Ruhr University Bochum, gregor.leander@rub.de
Dr. Christof Beierle	Research Associate (Faculty position), Symmetric Cryptography, Faculty of Computer Science, Ruhr University Bochum, christof.beierle@rub.de
Additional referee to be provided later	following my research visit to KU Leuven in April–May 2026, I expect to add a 1 further referee from there.

Awards and scholarships

2023	The Luxembourg Bankers' Association (ABBL) Talent grant.

Skills & Languages

Programming Languages	C/C++, CUDA, Python, SageMath, SQL, Bash
English	Advanced Knowledge (C1 Level)
German	Intermediate Knowledge (B1 Level, Goethe-Certificate)
French	Basic knowledge (A1-A2 Level)
Russian	Mother tongue
Ukrainian	Mother tongue

Links

Github
Linkedin

Bochum, April 8, 2026